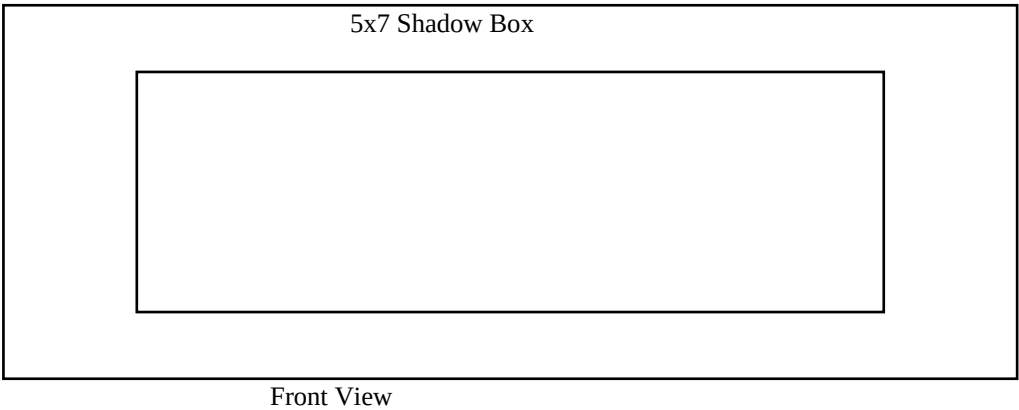


Living Landscape Shadow Box v0.1

Concept design document for OpenSCAD and Blender implementation.

Design Goals

Create a 5x7 landscape illuminated shadow box using a Waveshare ESP32-P4 5" display mounted behind a removable 3D printed silhouette panel. The display provides animated skies, waves and lighting while the printed silhouette provides physical foreground detail.



Layer Stack

Layer	Notes
Glass door	Existing hinged glass
3D printed silhouette	0.8 mm visible face, matte black PLA, 0.2 mm nozzle
Integrated rear ribs	~2 mm deep, hidden
Embedded magnets	3x1 mm neodymium in corner pods
Air gap	2–3 mm
5" ESP32-P4 display	Landscape orientation (rotated framebuffer)
3D printed carrier	PLA-CF or standard PLA
ESP32-P4 PCB	Mounted behind display

Mechanical Notes

- Frame: 5×7 hinged shadow box (4×6 interior).
- Removable 4×6 printed panel with full border.
- Four 1/4" × 1/4" mild-steel squares glued to inside corners of the door.
- Magnets embedded in hidden corner pods of printed panel.
- Hidden ribs only behind rocks/buildings/lighthouse to preserve a thin visible edge.
- Print visible panel face-down using matte black PLA and 0.2 mm nozzle.

Software Pipeline

1. Generate SVG silhouette.
2. Import into OpenSCAD for parametric border, ribs, magnet pockets.
3. Export STL.
4. Refine in Blender (optional sculpting of rocks/waves).
5. Slice in Bambu Studio (0.08 mm layers).
6. ESP32-P4 renders animated skies, waves, moon, beacon, weather.